

Seed 6, Round 2, Layer 4: the Ehrhart route to $S1$.
 Sharp cap $M_j = j - 1$, the deep-interior theorem, an unconditional
 low band $j \leq 48$, Harnack monotonicity, and a complete proof of
 $S1$ on the one-small-coordinate region for *all* j

Seed 6 (R2L4), Warnaar loop experiment

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Abstract

Everything is in the *true* (target-first) convention of `synthesis-layer3.md` §4(iv). We prove: (i) a sharp Cap-Compression theorem: $\text{har}_j(c)$ depends on c only through $(\min(c_i, j-1))_i$ (improving $M = 2j$ of G10 and the empirical $M = j$ of Seed 4), with a realization corollary making $S1@j$, $S2@j$ finitely decidable for each j ; (ii) the deep-interior theorem: $\min_i c_i \geq j-1$ implies $\text{har}_j(c) = \text{har}_j^\infty$, with closed forms, so $S1/S2$ hold at every such (j, c) for all j ; (iii) the computational theorem $[q^j]N_2 \geq 0$ for all profiles, all d with $\gcd(d, 3) = 1$, for all $j \leq 48$ (previously $j \leq 5$); (iv) $S1$ is *false* without the hypothesis $3 \nmid d$ ($\text{har}_{13}((1, 1, 1)) = -1$), so the residue hypothesis is essential; (v) a new structural conjecture (Harnack monotonicity, HM), finitely decidable per level, verified completely for $j \leq 30$, together with a proved reduction $\text{HM} \Rightarrow S1$; (vi) the main new theorem: on the region “at least two coordinates $\geq j-1$ ” (any third coordinate, any d), $\text{har}_j(c) \geq 0$ for all $j \notin \{2, 4\}$ — proved for *all* j simultaneously by an exact chamber quasi-polynomial computation (period 12, degree 3; the chamber walls are enumerated exactly from the switch loci of the defining sums — $2a = j$, $4a = j$, a , $j - a$, all with offsets ≤ 6 — and the pieces are pinned by full-rank exact interpolation, the $4a = j$ family carrying no jump), whose cubic part is the single global cubic $j^3\psi(a/j)$, $\psi(\mu) = (2\mu^3 - 6\mu^2 + 6\mu + 1)/24$, with $\psi'(\mu) = (\mu - 1)^2/4 \geq 0$, explicit quadratic slop $(5/8)j^2 + 7j + 4$, tail threshold $j \geq 23$, and an exact finite check below it. Together with (iii), the only part of $S1$ still open is: $j \geq 49$ with at least two coordinates of c smaller than $j - 1$.

1 Conventions and prior identities

Profiles are $c = (c_0, c_1, c_2) \in \mathbb{Z}_{\geq 0}^3$, $|c| = d$. Throughout

$$\text{EMD}(c, c') = 3 \max(0, c'_1 - c_1, c_0 - c'_0) + (c'_0 - c_0) - (c'_1 - c_1),$$

and the target-first kernel $(1 + q^m + q^{2m})H_{c,m} = \sum_{c'} q^{m \text{EMD}(c,c')} H_{c',m-1}$ (reference implementation `scripts/seed8_R2L3_engine.sage`). As in Layer 3 (rederived and machine-verified verbatim in the true convention; `seed6_R2L4_engine.py`, $d = 4, 5, 7$):

$$N_2(c) := (1 + q^2 + q^4) Q_{2,c} = [B_c(q^2) - 1 - q^2 - q^4] + \text{har}(c),$$

$$\text{har}(c) = \sum_{c' \neq c} q^{2 \text{EMD}(c,c')} Q_{1,c'} - q(1 + q + q^2 + q^3 + q^4) Q_{1,c}, \quad B_c(q) = \sum_{c' \in \Delta_d} q^{\text{EMD}(c,c')},$$

where $H_{1,c} = B_c(q) \frac{1-q}{1-q^3}$ and $Q_{1,c} = H_{1,c} - 1$. The ball term $B_c(q^2) - 1 - q^2 - q^4$ is coefficientwise ≥ 0 (L-ball, Layer 3). Write $\text{har}_j(c) := [q^j]\text{har}(c)$. The sub-conjectures of Layer 3 are

$$S1 : \text{har}_j(c) \geq 0 \quad (j \notin \{2, 4\}), \quad S2 : \text{har}_4(c) \geq -(b_2(c) - 1),$$

for $\gcd(d, 3) = 1$; $\text{har}_2(c) = -(b_1(c) - 1)$ exactly. Here $b_e(c) = \#\{c' \in \Delta_d : \text{EMD}(c, c') = e\}$.

1.1 Deviation geometry (true convention)

For $u = c' - c$ write $s = u_0, t = u_1$ ($u_2 = -s - t$). Then $\text{EMD}(c, c') = g(s, t) := 3 \max(0, t, -s) + s - t$, and $g \equiv s - t \pmod{3}$. The sphere $S_e = \{g = e\}$ ($e \geq 1$) is the boundary of the triangle with vertices $(e, 0), (-e, e), (0, -e)$ and has exactly $3e$ points; every $u \in S_e$ has $|u_0|, |u_1|, |u_2| \leq e$. The point $c + u$ is a valid profile iff $s \geq -c_0, t \geq -c_1, s + t \leq c_2$. EMD is invariant under the cyclic rotation $c \mapsto (c_1, c_2, c_0)$ applied to both arguments (machine-verified), so all statements below are rotation-invariant.

1.2 Local formula for har_j

With $A_k(c) = \#\{c' \in \Delta_d : g(c' - c) \leq k, g \equiv k \pmod{3}\} = \sum_{e \leq k, e \equiv k(3)} b_e(c)$ we have $[q^k]Q_{1,c} = A_k(c) - A_{k-1}(c) - [k=0]$, and, expanding the definition,

$$\text{har}_j(c) = \sum_{e=1}^{\lfloor (j-1)/2 \rfloor} \sum_{u \in S_e, c+u \text{ valid}} [q^{j-2e}]Q_{1,c+u} - \sum_{i=1}^5 [q^{j-i}]Q_{1,c}, \quad (1)$$

where $[q^k]Q_1 := 0$ for $k \leq 0$. (Machine-verified against the full polynomial engine at $d = 4, 5, 7$, all profiles, all j .)

2 The sharp cap $M_j = j - 1$ (Cap-Compression, sharp form)

Theorem 2.1 (CAP-SHARP). *For every $j \geq 1$, $\text{har}_j(c)$ depends on c only through $(\min(c_i, j - 1))_{i=0,1,2}$.*

Proof. Two facts. (F1) Every $u \in S_e$ has $|u_i| \leq e$ (checked edge by edge on the triangle). (F2) $[q^k]Q_{1,c'} = A_k(c') - A_{k-1}(c') - [k=0]$ counts lattice points at g -distance $\leq k$ from c' ; by (F1) applied to all spheres of radius $\leq k$, the validity constraints only probe whether each c'_i is $\geq$ some threshold $\leq k$; hence $[q^k]Q_{1,c'}$ depends only on $(\min(c'_i, k))_i$.

Suppose $\min(c_i, j - 1) = \min(\hat{c}_i, j - 1)$ for all i ; then for each i either $c_i = \hat{c}_i \leq j - 2$ or both $c_i, \hat{c}_i \geq j - 1$. In (1): (1) validity of $u \in S_e$ probes thresholds $\leq e \leq \lfloor (j - 1)/2 \rfloor \leq j - 1$, so agrees for $c, \hat{c}$; (2) for a neighbour term with $k = j - 2e$: if $c_i \geq j - 1$ then $(c + u)_i \geq j - 1 - e \geq j - 2e = k$ (as $e \geq 1$), so $\min((c + u)_i, k) = k = \min((\hat{c} + u)_i, k)$; if $c_i = \hat{c}_i$ the two agree trivially; (3) the self terms probe thresholds $j - i \leq j - 1$. By (F2) every term agrees. $\square$

Corollary 2.2 (realization; finite decidability of $S1@j$). *Every capped class $(\min(c_i, j - 1))_i$ arising from a profile with $\gcd(|c|, 3) = 1$ is realized by a profile with all $c_i \leq j + 1$ and $\gcd(d, 3) = 1$: if some coordinate is $\geq j - 1$, choose it in $\{j - 1, j, j + 1\}$ to adjust d modulo 3; if all $c_i \leq j - 2$ the profile is its own representative. Hence an exact check of $S1@j$ (resp. $S2@j$) over the finitely many profiles with $c_i \leq j + 1, 3 \nmid d$, proves it for all d with $\gcd(d, 3) = 1$.*

3 The deep-interior theorem (region R0)

Theorem 3.1 (R0). *If $\min_i c_i \geq j - 1$ then $\text{har}_j(c) = \text{har}_j^\infty$, where with $E = \lfloor (j - 1)/2 \rfloor$,*

$$\text{har}_j^\infty = \sum_{e=1}^E 3e(j - 2e + 1) - \sum_{i=1}^5 u_{j-i}, \quad u_k = k + 1 \ (k \geq 1), \ u_k = 0 \ (k \leq 0),$$

i.e. $\text{har}_{0..5}^\infty = 0, 0, -2, 1, 0, 10$ and, for $j \geq 6$,

$$\text{har}_{2p}^\infty = \frac{(p-1)(2p^2+5p-20)}{2}, \quad \text{har}_{2p+1}^\infty = p(p+1)(p+2) - 10p + 5.$$

In particular $\text{har}_j^\infty \geq 0$ for all $j \neq 2$, and $\text{har}_2^\infty = -2 = -(b_1 - 1)$ is absorbed by the ball term. Hence S1 and S2 hold at every pair (j, c) with $\min_i c_i \geq j - 1$ (all j , all d).

Proof. By Theorem 2.1 (same bookkeeping) all validity constraints in (1) are inactive: every sphere is full, $b_e = 3e$, so $A_k = \sum_{e \leq k, e \equiv k(3)} 3e$ gives $[q^k]Q_1^\infty = u_k = k + 1$ for $k \geq 1$ (a three-case computation on $k \bmod 3$), and each of the $3e$ neighbours at distance e contributes $u_{j-2e} = j - 2e + 1$. The closed forms are elementary summation. Machine-verified for $j \leq 29$ at $c = (60, 58, 60)$ against the full engine. $\square$

4 Unconditional low band: $j \leq 48$

Theorem 4.1 (computational; `seed6_R2L4_sweep.py`). *For every $j \leq 48$: S1@ j holds ($j \notin \{2, 4\}$), $\text{har}_2 = -(b_1 - 1)$ exactly, and S2@4 holds — for all profiles c and all d with $\gcd(d, 3) = 1$. Consequently $[q^j]N_2 \geq 0$ for all $j \leq 48$, all such d .*

Proof. By Corollary 2.2 each level j reduces to the finite set $\{c : c_i \leq j + 1, 3 \nmid |c|\}$ (up to rotation); the sweep evaluates (1) there in exact integer arithmetic. All levels pass; the minimum margin is 0, attained at $c = (0, 0, 1)$ at every level. The sweep's har was cross-checked against the full polynomial engine for $d \in \{4, 5, 7, 8, 10\}$, $j \leq 25$, all profiles. $\square$

5 $3 \mid d$: the hypothesis is essential

Remark 5.1. $\text{har}_j(c)$ is defined by (1) for every $c \in \mathbb{Z}_{\geq 0}^3$ regardless of $d \bmod 3$. As a pure lattice statement S1 fails at $3 \mid d$: e.g. $\text{har}_{13}((1, 1, 1)) = -1$ and $\text{har}_{15}((0, 1, 2)) = -1$ (`seed6_R2L4_mono.py`); all observed failures have $3 \mid d$. Any correct proof of S1 must therefore use $\gcd(d, 3) = 1$.

6 Harnack monotonicity (HM): a finitely decidable master reduction

Conjecture 6.1 (HM). *Let $j \notin \{2, 4\}$. Then (M1) if $|c| \equiv 1 \pmod{3}$: $\text{har}_j(c + e_i) \geq \text{har}_j(c)$ for $i = 0, 1, 2$; (M2) if $|c| \equiv 2 \pmod{3}$: $\text{har}_j(c + e_i + e_k) \geq \text{har}_j(c)$ for all $i \leq k$.*

Proposition 6.2 (base case). *At $d = 1$, $B_c = 1 + q + q^2$, so $H_1 = 1$, $Q_1 = 0$ and $\text{har}_j \equiv 0$ for all j .*

Theorem 6.3 (reduction). *HM $\Rightarrow$ S1 (all j , all c , $\gcd(d, 3) = 1$). Moreover HM@ j is finitely decidable for each j (Corollary 2.2: the box $c_i \leq j + 1$ suffices, steps above the cap being automatically tight), and has been verified completely for all $j \leq 30$ (0 failures; $j \leq 18$: 158,652 steps, 38,295 tight; $j = 19..30$: `seed6_R2L4_mono_hi.py`).*

Proof. Any c with $3 \nmid d$ walks down to $d = 1$ by residue-respecting steps ($d \equiv 2$: remove one box; $d \equiv 1, d \geq 4$: remove two), staying in $3 \nmid d$. By HM, har_j is non-increasing along the reversed walk and vanishes at $d = 1$ by Proposition 6.2. $\square$

Remark 6.4. HM is the recommended structural target: it is local (one or two boxes), all failures of its naive (non-residue-respecting) form occur exactly on steps touching $3 \mid d$ — it “knows” where the hypothesis enters (Remark 5.1).

7 Main theorem: $S1$ on the one-small-coordinate region, all j

By rotation invariance assume $c = (c_0, c_1, a)$ with $c_0, c_1 \geq j - 1$, $a \geq 0$ arbitrary. By Theorem 2.1, $\text{har}_j(c)$ depends only on $\min(a, j - 1)$; for $a \geq j - 1$ Theorem 3.1 applies. So fix $0 \leq a \leq j - 2$ and set $\varphi_j(a) := \text{har}_j(c)$ (any d , no residue condition).

7.1 The exact formula for φ

Lemma 7.1 (ingredients; also machine-verified in `seed6_R2L4_r1.py`). *Let $e \geq 1$, $y(m) = \lceil m/2 \rceil$ for $m \geq 1$ (0 else). Then:*

1. $\mu_e(v) := \#\{u \in S_e : u_0 + u_1 = v\} = 1 + [v \equiv e \pmod{2}] - [|v| = e]$ for $|v| \leq e$, 0 otherwise.
2. $\#\{u \in S_e : u_0 + u_1 > \beta\} = \lfloor (3\delta - 1)/2 \rfloor$ for $\delta = e - \beta \geq 1$ (0 else), $\beta \geq 0$.
3. $x(m) := \sum_{r=0}^{\lfloor (m-1)/3 \rfloor} \lfloor (3(m-3r)-1)/2 \rfloor = \lfloor (m+1)^2/4 \rfloor$ and $y(m) = x(m) - x(m-1) = \lceil m/2 \rceil$.
4. $[q^k]Q_1$ of a profile (∞, ∞, β) (only the third constraint active) equals $w_k(\beta) = (k+1) - y(k-\beta)$ for $k \geq 1$.

Proof. Parametrize the three edges of S_e (in the $(s, t) = (u_0, u_1)$ plane, §1) explicitly: $\mathcal{E}_1 = \{(s, s - e) : 0 \leq s \leq e\}$ (the branch $\max = 0$ of g), $\mathcal{E}_2 = \{(e - 2t, t) : 0 \leq t \leq e\}$ ($\max = t$), and $\mathcal{E}_3 = \{(s, -2s - e) : -e \leq s \leq 0\}$ ($\max = -s$), overlapping exactly at the three vertices $(e, 0)$, $(-e, e)$, $(0, -e)$.

(1) The height $v = u_0 + u_1$ equals $2s - e$ on $\mathcal{E}_1$ (each value $v \equiv e \pmod{2}$ with $|v| \leq e$, exactly once), $e - t$ on $\mathcal{E}_2$ (each $0 \leq v \leq e$ once), and $-s - e$ on $\mathcal{E}_3$ (each $-e \leq v \leq 0$ once); the shared vertices are double-counted at $v = e$, $v = 0$, $v = -e$ respectively. Hence, for $|v| \leq e$,

$$\mu_e(v) = [v \equiv e \pmod{2}] + [v \geq 0] + [v \leq 0] - [v=0] - [|v|=e] = 1 + [v \equiv e \pmod{2}] - [|v|=e].$$

(2) Sum (1) over $v = \beta + 1, \dots, e$ (since $\beta \geq 0$, the correction at $v = -e$ is not met): with $\delta = e - \beta \geq 1$ the sum is $\delta + \#\{\beta < v \leq e : v \equiv e \pmod{2}\} - 1 = \delta + \lceil \delta/2 \rceil - 1 = \lfloor (3\delta - 1)/2 \rfloor$ (check the two parities of δ).

(3) The shift identity $x(m) - x(m-6) = 3(m-2)$ for $m \geq 7$, plus six base cases.

(4) With the single constraint $u_0 + u_1 \leq \beta$, part (2) gives $b_e = 3e - \lfloor (3(e-\beta) - 1)/2 \rfloor$, so $A_k - A_{k-1}$ differs from the free value $u_k = k + 1$ of Theorem 3.1 by $\sum_{e \leq k, e \equiv k(3)} \lfloor \frac{3(e-\beta)-1}{2} \rfloor - \sum_{e \leq k-1, e \equiv k-1(3)} \lfloor \frac{3(e-\beta)-1}{2} \rfloor = x(k-\beta) - x(k-1-\beta) = y(k-\beta)$ by (3). $\square$

Proposition 7.2 (φ formula; proved by the CAP-SHARP bookkeeping, machine-verified against the engine for $j \leq 33$, all a). *With $E = \lfloor (j-1)/2 \rfloor$ and $n_e = j - 2e - a$,*

$$\varphi_j(a) = \text{har}_j^\infty - C_1 - C_2 + C_3,$$

$$C_1 = \sum_{e=a+1}^E (j-2e+1) \left\lfloor \frac{3(e-a)-1}{2} \right\rfloor, \quad C_2 = \sum_{e=1}^E \sum_{v=\max(-e, 1-n_e)}^{\min(e, a)} \mu_e(v) y(n_e+v), \quad C_3 = \sum_{i=1}^5 y(j-a-i).$$

(C_1 : neighbours pushed outside the half-space $u_0 + u_1 \leq a$ lose their entire contribution $u_{j-2e} = j - 2e + 1$; C_2 : surviving neighbours at signed height v have their own count reduced by $y(n_e + v)$ by Lemma 7.1(4); C_3 : the self terms.)

The inner v -sum of C_2 closes in $O(1)$ using $\sum_{m=1}^M \lceil m/2 \rceil = \lfloor (M+1)^2/4 \rfloor$ and its parity refinements, giving an $O(j)$ exact evaluator (`seed6_R2L4_r1poly.py`; verified identical to the double sum for $j \leq 60$, all a).

7.2 Chamber quasi-polynomials

Lemma 7.3 (structure). *Define the wall list*

$$\mathcal{W} = \{2a = j+t : |t| \leq 2\} \cup \{4a = j+t : |t| \leq 2\} \cup \{a = t : 0 \leq t \leq 2\} \cup \{a = j-t : 2 \leq t \leq 6\}.$$

All pairwise intersections of distinct walls of $\mathcal{W}$ have $j \leq 14$. Every integer point of $\{j \geq 15, 0 \leq a \leq j-2\}$ lies on a wall of $\mathcal{W}$ or in one of the three convex, wall-free regions

$$\text{LOW}_A = \{4a < j-2, a > 2\}, \quad \text{LOW}_B = \{4a > j+2, 2a < j-2\}, \quad \text{HIGH}' = \{2a > j+2, a < j-6\},$$

and, restricted to a residue class $(j, a) \bmod 12$:

1. *on each of $\text{LOW}_A, \text{LOW}_B, \text{HIGH}'$, φ agrees with a polynomial in (j, a) of degree ≤ 3 ;*
2. *on each wall line of $\mathcal{W}$, φ agrees with a quasi-polynomial of degree ≤ 3 and period 12 in the line parameter.*

($\mathcal{W}$ is an over-approximation: a listed line need not carry an actual jump. Theorem 7.4 shows that the $4a = j+t$ family, the lines $a = t$, and $a = j-t$ with $4 \leq t \leq 6$ carry none.)

Proof. We enumerate the switch loci of the four constituents of Proposition 7.2 exactly.

har_j^∞ is a quasi-polynomial in j of degree 3, period 2: no walls.

C_3 . Since $\lceil m/2 \rceil = 0$ for $m \in \{0, -1\}$, the truncation in y is invisible whenever $m \geq -1$; and $j - a - i \geq -1$ for all $1 \leq i \leq 5$ iff $a \leq j - 4$. So $C_3 = \sum_{i=1}^5 \lceil (j - a - i)/2 \rceil$ (degree 1, period 2) on $a \leq j - 4$; the only switch loci are $a = j - 2, j - 3$.

C_1 . Substituting $e = a + \delta$, $C_1 = \sum_{\delta=1}^{E-a} (j - 2a - 2\delta + 1) \lfloor (3\delta - 1)/2 \rfloor$: the summand is a quasi-polynomial of degree 2, period 2 in δ , and the sum is empty iff $a \geq E$. On a residue class mod 12 the upper limit $E - a$ is a linear form of fixed parity, so closed-form summation gives a polynomial of degree ≤ 3 ; the only switch locus is $a = E$, i.e. $2a = j + t, t \in \{-1, -2\}$.

C_2 . Write $n = n_e$, $U = \min(e, a)$, $m_1 = \max(1, n - e)$, $m_2 = n + U$, $p = (n + e) \bmod 2$, $G(M) = \sum_{m=1}^M \lceil m/2 \rceil = \lfloor (M+1)^2/4 \rfloor$, and G_p its restriction to $m \equiv p \pmod{2}$. Every surviving $m = n_e + v$ satisfies $m \geq m_1 \geq 1$, so expanding $\mu_e(v) = 1 + [v \equiv e \pmod{2}] - [v=e] - [v=-e]$ (Lemma 7.1(1)) closes the inner sum:

$$\sum_v \mu_e(v) y(n+v) = G(m_2) - G(m_1-1) + G_p(m_2) - G_p(m_1-1) - [n-e \geq 1] y(n-e) - [e \leq a] y(n+e).$$

As a function of $e \in [1, E]$ this is a quasi-polynomial of degree ≤ 2 , period 2, on each interval cut out by the breakpoints

$$B_1 : e = a \quad (\text{from } U \text{ and } [e \leq a]), \quad B_2 : 3e = j - a + t, \quad t \in \{0, -1\} \quad (\text{from } m_1 \text{ and } [n - e \geq 1]),$$

$B_3 : e = j - a - 1$ (emptiness of the branch $e \leq a$, where $m_2 = j - a - e$; for $e > a$, $m_2 = j - 2e \geq 1$ automatically), together with the range ends $B_4 : e = E$ and $B_5 : e = 1$. On a fixed residue class mod 12 every breakpoint is a linear form in (j, a) (denominators dividing 6), so summation over e yields a polynomial of degree ≤ 3 in (j, a) on every region where the order of the breakpoints, clamped to $[1, E]$, is constant. That order changes exactly on the pairwise collision loci: $B_1=B_2: 4a = j + t, t \in \{0, -1\}$; $B_1=B_3, B_1=B_4, B_3=B_4: 2a = j + t, -2 \leq t \leq 0$; $B_2=B_5: a = j - t, t \in \{3, 4\}$; $B_3=B_5: a = j - 2$; $B_1=B_5$, and emptiness of the branch $e \leq a: a \in \{0, 1\}$; $B_2=B_4: a \leq 3 - \lfloor j/2 \rfloor < 0$ for $j \geq 8$, outside the domain; $B_4=B_5: j \leq 4$.

Collecting all loci gives a subset of $\mathcal{W}$; the degree bound is the counting just performed, and the period divides 12 (lcm of the endpoint denominators 2, 3 with the parity period 2). Every pairwise intersection of distinct walls of $\mathcal{W}$ has $j \leq 14$ (largest case: $2a = j + t$ meets $a = j - t'$ at $j = t + 2t' \leq 14$), and one checks directly that none of the walls of $\mathcal{W}$ meets LOW_A , LOW_B or HIGH' in $\{j \geq 15\}$; the three regions are convex, hence each is contained in a single order-pattern region, proving (1). For (2), substitute the line parametrizations ($a = s, j = 2s - t$; $a = s, j = 4s - t$; $a = t, j = s$; $a = s, j = s + t$) into the same enumeration: all breakpoints become linear forms in s with denominators dividing 6, and their collisions occur only at wall-wall intersections, i.e. at $j \leq 14$; hence for $j \geq 15$ the restriction of φ to each wall line is a quasi-polynomial of degree ≤ 3 and period 12 in s . $\square$

Theorem 7.4 (chamber polynomials; `seed6_R2L4_r1poly.py` and `repair6L4_walls.py`, exact rational arithmetic). *There are polynomials $P_r^{\text{LOW}}, P_r^{\text{HIGH}}$ ($r = (j, a) \bmod 12$, 144 classes each, degree ≤ 3) and $P_{j \bmod 12}^{(n)}$ ($n = 2, 3$, cubics in j) such that for all $j \geq 6$, $0 \leq a \leq j - 2$,*

$$\varphi_j(a) = \begin{cases} P_{j \bmod 12}^{(j-a)}(j), & a \in \{j - 2, j - 3\}, \\ P_{(j,a) \bmod 12}^{\text{LOW}}(j, a), & 2a < j, \\ P_{(j,a) \bmod 12}^{\text{HIGH}}(j, a), & 2a \geq j, a \leq j - 4. \end{cases}$$

In particular the candidate walls $4a = j + t$ of Lemma 7.3 carry no polynomial jump. Moreover:

1. $P_r^{\text{LOW}} = P_r^{\text{HIGH}}$ whenever j is odd (the jump across $2a = j$ exists only for even j);
2. every one of the 288 chamber polynomials has cubic part

$$\frac{j^3 + 6j^2a - 6ja^2 + 2a^3}{24} = j^3 \psi(a/j), \quad \psi(\mu) = \frac{2\mu^3 - 6\mu^2 + 6\mu + 1}{24},$$

and both strip families have leading coefficient $\frac{1}{8} = \psi(1)$;

3. *the total absolute size of the sub-cubic coefficients is bounded, over all classes and chambers, by $K_2 = \frac{5}{8}$ (degree 2), $K_1 = 7$ (degree 1), $K_0 = 4$ (constants).*

Proof. By Lemma 7.3 it suffices to identify the polynomial on each region and each wall line: a polynomial of degree ≤ 3 in (j, a) is determined by any full-rank set of 10 interpolation conditions, and a cubic on a line by 4 points. All of the following is exact (integer/rational arithmetic), with 0 mismatches.

- *Regions* (`repair6L4_walls.py`, step 2): for each residue class, the point sets $\{4a \leq j - 8, a \geq 3\} \subset \text{LOW}_A$, $\{4a \geq j + 8, 2a \leq j - 8\} \subset \text{LOW}_B$, and $\{2a \geq j + 8, a \leq j - 9\} \subset \text{HIGH}'$, with $15 \leq j < 620$ (45,300 / 44,551 / 88,506 points), give full-rank (rank 10), consistent linear systems whose unique solutions equal P^{LOW} , P^{LOW} , P^{HIGH} respectively; in particular the two sides of the $4a = j$ family carry the *same* polynomial.

- *Wall lines* (`repair6L4.walls.py`, step 3): on every line $2a - j = t$, $4a - j = t$ ($|t| \leq 8$), $a = t$ ($0 \leq t \leq 8$), $j - a = t$ ($2 \leq t \leq 8$) — a superset of $\mathcal{W}$ — and every residue class of the line parameter mod 12, the displayed formula matches φ at 10 points with $j \geq 15$ (6,000 points in all; 4 per class already pin the quasi-polynomial of Lemma 7.3(2)).
- *Head and safety net* (`seed6_R2L4_r1poly.py` pipeline): dense band $6 \leq j < 150$, all $0 \leq a \leq j - 2$ (11,016 points — covering in particular every $j \leq 14$); strips $6 \leq j < 400$; sparse band to $j = 610$; sliver lines $|2a - j| \leq 30$, $j - a \leq 30$, $a \leq 30$ for j up to 800 (19,639 points).

By Lemma 7.3, every integer point with $j \geq 15$ lies in one of the three regions or on a wall line of $\mathcal{W}$; in either case φ agrees there with a (quasi-)polynomial that the verified points pin to the displayed one. Points with $6 \leq j \leq 14$ are verified directly in the dense band. Items (1)–(3) are read off the exact fitted coefficients. $\square$

Theorem 7.5 (R1: main). *Let $j \notin \{2, 4\}$ and let c be any profile with at least two coordinates $\geq j - 1$ (the third arbitrary, any d). Then $\text{har}_j(c) \geq 0$; indeed > 0 for $j \geq 5$.*

Proof. By the reductions above it suffices that $\varphi_j(a) \geq 0$ for $0 \leq a \leq j - 2$. *Tail.* $\psi'(\mu) = (\mu - 1)^2/4 \geq 0$, so $\psi \geq \psi(0) = \frac{1}{24}$ on $[0, 1]$. By Theorem 7.4(2,3), for (j, a) in LOW/HIGH,

$$\varphi_j(a) \geq \frac{j^3}{24} - \left(\frac{5}{8}j^2 + 7j + 4 \right) > 0 \quad (j \geq 23),$$

using $0 \leq a \leq j$. The strip cubics are positive for $j \geq 11$ (same comparison with leading coefficient $\frac{1}{8}$; they are in fact positive from $j = 10$ by direct evaluation, and the head check covers $j \leq 22$). *Head.* Exact evaluation for $j \leq 22$, $j \notin \{2, 4\}$, all $0 \leq a \leq j - 2$:

$$\min_a \varphi_j(a) = 0, 0, 0 \quad (j = 0, 1, 3); \quad 3, 4, 12, 17, 31, 41, 61, 76, 104, 126, 162, 191, 237, 275, 331, 378, 446, 504 \quad (j = 5..22).$$

All ≥ 0 . (For the excluded exponents, on the domain $0 \leq a \leq j - 2$: $\min_a \varphi_2 = \varphi_2(0) = -1$ and $\min_a \varphi_4 = -1$, exactly the known ball-absorbed cases; the value $\text{har}_2^\infty = -2$ is φ_2 at $a = j - 1$, which belongs to R0.) $\square$

Corollary 7.6. *$S1$ holds (all j simultaneously) on all profiles with at most one coordinate $< j - 1$. Combined with Theorem 4.1, the only remaining open part of $S1$ is: $j \geq 49$ and at least two coordinates of c less than $j - 1$.*

8 Status, lift to level n , and open regions

Proved here (all d , $\gcd(d, 3) = 1$ **where relevant**). Theorem 2.1 (sharp cap $M_j = j - 1$) and Corollary 2.2 (finite decidability per level — the explicit $J_0 \rightarrow \infty$ mechanism); Theorem 3.1 (R0, all j); Theorem 4.1 ($j \leq 48$ unconditional); Proposition 6.2 and Theorem 6.3 ($\text{HM} \Rightarrow S1$; $\text{HM}@j$ proved for $j \leq 30$); Theorem 7.5 (R1, all j). Since $S2$ concerns only the single exponent $j = 4 \leq 48$, Theorem 4.1 makes $S2$ *fully proved, unconditionally*.

Verified (not yet proved). Conjecture 6.1 (HM) for $j \geq 31$: open (finitely decidable per level). The remaining open region of $S1$ (Corollary 7.6) is R2/R3: two or three coordinates $< j - 1$ with $j \geq 49$; by Theorem 2.1 it is, per level, a finite question in the box $c_i \leq j + 1$.

Lift to n . Unchanged from Layer 3 (Y4): the Sphere Absorption pattern is verified at $n = 2$ for $d \leq 35$ (now superseded on the low band by Theorem 4.1) and at $n = 3$ for $d = 4, 5$. The natural generalization of the present R0/R1 analysis to level n (deviation spheres under q^n^{EMD} with Q_{n-1} -weights) is set up by the level- n analogue of (1) but was not pursued here.

Artifacts. Scripts (exact integer/rational arithmetic): `scratch/scripts/seed6_R2L4_engine.py` (engine + T1/T2 + local formula), `seed6_R2L4_sweep.py` (+ log; Theorem 4.1), `seed6_R2L4_mono.py`, `seed6_R2L4_mono2.py`, `seed6_R2L4_mono_hi.py` (+ log; HM), `seed6_R2L4_r1.py` (Lemma 7.1, Proposition 7.2), `seed6_R2L4_r1poly.py` (self-contained certificate of Theorems 7.4 and 7.5: run with argument `pipeline`); `repair6L4_walls.py` (+ log; wall-locus pinning certificate for Lemma 7.3 and Theorem 7.4: sub-chamber full-rank fits and wall-line verification); notes `scratch/prove-seed6-layer4.md` and `scratch/repair-seed6-layer4.md`.